ETALE MORPHISMS OF SCHEMES ARE EXACTLY THE LOCALLY FINITELY PRESENTED FORMALLY ETALE ONES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1705-1714 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 1716-1742, the complete author proof (25 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Formally etale morphisms, the decomposition of etaleness into flat, unramified and locally finitely presented, and the algebra characterisation of formally etale ring maps; the converse is retained.

Source reference. Original lemma 00UR: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1742–1742. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-affine-formally-etale: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1655–1661. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-formally-etale-on-opens: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1574–1578. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-unramified-flat-formally-etale: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1626–1628. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02GJ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1742–1742. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01TQ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1742–1742. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02GH: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1742–1742. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. The following are equivalent:*

(1) *The morphism f is étale, and*
(2) *the morphism f is locally of finite presentation and formally étale.*

Proof. Assume f is étale. An étale morphism is locally of finite presentation, flat and unramified, see Morphisms, Section 02GH. Hence f is locally of finite presentation and formally étale, see Lemma lemma-unramified-flat-formally-etale.

Conversely, suppose that f is locally of finite presentation and formally étale. Being étale is local in the Zariski topology on X and S, see Morphisms, Lemma 02GJ. By Lemma lemma-formally-etale-on-opens we can cover X by affine opens U which map into affine opens V such that $U \to V$ is formally étale (and of finite presentation, see Morphisms, Lemma 01TQ). By Lemma lemma-affine-formally-etale we see that the ring maps $\mathcal{O}(V) \to \mathcal{O}(U)$ are formally étale (and of finite presentation). We win by Algebra, Lemma 00UR. (We will give another proof of this implication when we discuss formally smooth morphisms.) □